

Web Application “Bank”

This project is a web application for a bank. It allows clients to manage accounts, deposits, loans, transfers, and currency exchange. The system also has manager and admin roles. Managers process client requests, and admins manage bank settings. The main goal of the project is to simulate basic online banking functionality.

User roles in the system:

Client can:

- open bank accounts in different currencies
- transfer money between own accounts
- transfer money to other clients by phone number or card number
- open deposits
- apply for loans
- exchange currencies
- view personal accounts, deposits, loans, and transactions
- change personal contact information, such as phone number and email

Manager can:

- approve or reject requests to open a bank account
- approve or reject requests to open a deposit
- review loan applications
- create loan offers for clients
- approve or reject loan requests
- calculate loan conditions, such as amount, rate, duration, and monthly payment

Admin can:

- update bank fees;
- update deposit rates;
- update loan rates;
- update currency exchange rates;
- update account limits;
- approve or reject manager registration requests.

Statuses of users:

- Active: can perform actions fully
- Deactivated: cannot perform actions, but can be activated later
- Deleted: cannot perform actions, and cannot be activated later

Accounts

To open a new account, the client fills in a request form. In this form, the client chooses:

- account name
- currency
- transaction limit

After this, the account request gets the Pending status.

The manager can approve or reject the request. If the request is approved, the account becomes active. If the request is rejected, the account gets the Rejected status.

The system also has a rule: one client can have no more than 10 active accounts. If the client already has 10 active accounts, the new account request can be rejected.

Each account stores important information:

- account owner;
- account name;
- card number;
- CVV code;
- expiry date;
- balance;
- currency;
- transaction limit;
- account status;
- information about whether this account is the main account.

Account statuses

- **Pending:** the account request was created by the client and is waiting for manager approval. The client cannot fully use this account yet.
- **Active:** the account is approved and works correctly. The client can use this account for transfers, currency exchange, deposits, loan payments, and other operations.
- **Deactivated:** the account was turned off by the client. It cannot be used for normal operations. If the deactivated account was the main account, the system removes it as the main account.

- **Expired:** the account is expired. It cannot be used anymore and cannot be activated again.
- **Deleted:** the account is deleted from active use. It cannot be used for operations and cannot be activated again.
- **Rejected:** the manager rejected the request to open this account. Also, the system can reject a new account if the client already has the maximum number of active accounts.

Operations

- top up an account
- transfer money between own accounts
- transfer money to another client by phone number
- transfer money by card number
- transfer money to an external card
- exchange money between accounts with different currencies
- pay loan payments
- top up deposits from an account
- withdraw money from deposits to an account
- view account transactions

If accounts use different currencies, the system applies the current exchange rate during the operation.

Transaction Types

The application stores different types of transactions:

- Account top-up: money is added to a client account.
- Transfer between own accounts: money is transferred between accounts of the same client.
- Phone transfer: money is sent to another client by phone number.
- Card transfer: money is sent to another bank client by card number.
- External card transfer: money is sent to a card outside the bank.
- Currency exchange: money is exchanged between accounts with different currencies.
- Deposit opening: money is used to open a new deposit.
- Deposit top-up: money is added to an active deposit.
- Deposit withdrawal: money is withdrawn from a deposit to an account.
- Deposit interest: interest is accrued for an active deposit.
- Loan payment: money is used to pay a loan.

Deposits

To open a deposit, the client chooses:

- source account
- deposit product
- deposit conditions
- opening amount
- auto-renewal option if it is available
- reinvest interest option

Each deposit stores information:

- deposit owner;
- deposit type;
- current amount;
- currency;
- interest rate;
- duration;
- start date;
- auto-renewal status;
- reinvest interest status;
- deposit status.

Deposit Types:

Kopilka: it is useful for clients who want to save money, but also want to have access to their deposit.

The minimum amount starts from 10 dollars or an equivalent amount in another currency. The duration is from 6 to 36 months. The interest rate depends on the selected currency.

Strategy: it is a deposit for clients who want to save money with a higher rate. It is less flexible than Kopilka, because the client can top up the deposit, but cannot withdraw money before the end of the term.

Strategy does not allow partial withdrawal. The minimum amount starts from 100 dollars or an equivalent amount in another currency. The duration is from 3 to 24 months. The interest rate is higher than for Kopilka.

Capital: is a long-term deposit for clients who want to keep money and receive the highest rate. This deposit is more strict than the other products.

The minimum amount starts from 1000 dollars or an equivalent amount in another currency. The duration is from 3 to 36 months. The interest rate is usually higher than for Strategy.

Managers can perform deposit operations:

- approve deposit requests;
- reject deposit requests;
- accrue interest for active deposits;

- process expired deposits.

Deposit Statuses

- **Pending:** the deposit request was created by the client and is waiting for manager approval. The deposit is not active yet.
- **Active:** the deposit was approved and works correctly. The system can accrue interest for this deposit.
- **Expired:** the deposit term is finished. The deposit stops working if auto-renewal is not enabled.
- **Deleted:** the deposit is deleted from active use.
- **Rejected:** the manager rejected the request to open this deposit.
- **Closed:** the deposit is closed and is no longer active.

Loans

Personal Loan

A general-purpose loan that can be used for any needs

- Amount: from 50,000 to 5,000,000 KZT
- Duration: from 3 to 60 months
- Interest rate: from 21% annually

Car Loan

A loan intended for purchasing new or used vehicles.

- Amount: from 500,000 to 20,000,000 KZT
- Duration: from 12 to 84 months
- Interest rate: from 18% annually

Mortgage Loan

A loan intended for purchasing real estate.

- Amount: from 3,000,000 to 80,000,000 KZT
- Duration: from 3 to 20 years
- Interest rate: from 16% annually

Loan Application Process

1. A client selects a loan type and submits a loan request with the desired amount
2. The request is sent to bank managers for review
3. A manager may:
 - reject the request
 - create one or more loan offers with different amounts, interest rates, and durations
4. The offers are sent to the client
5. The client may accept one of the offers or decline all offers
6. After acceptance, the loan becomes active

The loan interest rate, monthly payment amount, and payment schedule are determined by the manager when creating the offer

Loan Statuses

- **Pending:** loan application is waiting for manager review.
- **Offered:** one or more loan offers have been prepared and sent to the client.
- **Rejected:** loan application was rejected by the manager.
- **Refused:** the client declined the offered loan.
- **Active:** the loan is currently active and being repaid.
- **Closed:** the loan has been fully repaid.

Currency exchange:

Exchange Rates

Admin is responsible for updating currency exchange rates on a daily basis. The latest exchange rates are available to all clients and can be viewed at any time.

Currency Exchange Calculator

Clients can use a built-in currency exchange calculator to estimate the amount they will receive when converting one currency to another using the current exchange rates.

Currency Exchange Process

1. A client selects the source and target currencies.
2. The system calculates the exchange result based on the current exchange rate.
3. The client confirms the operation.
4. Funds are withdrawn from the source account and credited to the target account.

All completed currency exchange operations are recorded in the transaction history and can be viewed by the client.

Settings:

Language settings

Clients can change the application language at any time. The following languages are supported:

- English
- Russian
- Kazakh

The selected language is applied throughout the entire user interface.

Contact Information

Clients can update their contact details directly from the settings page.

Supported actions:

- Change phone number
- Change email address

Account Management

Clients can manage their account access through the settings section.

Available actions:

- Change password
- Log out of the application
- Delete their account

Optional (future) features:

- QR code and biometric login for faster and safer authentication.
- Automatic currency exchange rates from an external financial service.
- Loyalty program with bonuses, better deposit rates, or lower loan rates for active clients.
- Broker account for buying and managing company shares.
- Children accounts for parents who want to control and manage accounts for their children.